Math Midterm · Practice Exam 1

Exponents · Polynomials and factoring · Quadratic equations and parabolas · Statistics

Name: _____ Date: _____ Period: _____

Time: 60 minutes · 80 points. Part A: 18 multiple-choice questions, 2 points each (36 points). Circle the letter of the best answer. Part B: 5 free-response questions (44 points). Show all your work in the space given: a correct answer with no work earns at most 1 point. "Exact" means leave answers as fractions and simplified radicals. A scientific or graphing calculator may be used; no formula sheet is provided.

Standard normal table (area to the LEFT of z). Use these values where a question says "use the table".

z	−2.00	−1.50	−1.00	−0.50	0.50	1.00	1.28	1.29	1.50	1.75	2.00
area	0.0228	0.0668	0.1587	0.3085	0.6915	0.8413	0.8997	0.9015	0.9332	0.9599	0.9772

Part A · Multiple Choice (2 points each, 36 points)

1. Simplify $\dfrac{(3x^2)^3}{9x^4}$.

A $\dfrac{x^2}{3}$ **B** $3x$

C $3x^2$ **D** $3x^{3/2}$

2. Evaluate $16^{-3/4}$.

A -8 **B** $\dfrac{1}{8}$

C -12 **D** $\dfrac{1}{12}$

3. Solve $3^{2x-1} = 27$.

A $x = 14$ **B** $x = 5$

C $x = \dfrac{3}{2}$ **D** $x = 2$

4. What is the remainder when $p(x) = 2x^3 - x^2 + 3x - 5$ is divided by $x - 2$?

A 13 **B** -31

C 21 **D** 0

5. For what value of k is $(x + 1)$ a factor of $x^3 + kx^2 + 2x - 4$?

A 1 **B** -7

C 7 **D** 3

6. Expand $(2x - 3)^2$.

A $4x^2 + 9$ **B** $4x^2 - 6x + 9$

C $2x^2 - 12x + 9$ **D** $4x^2 - 12x + 9$

7. Factor completely: $2x^2 - 18$.

A $2(x+3)(x-3)$ **B** $2(x^2-9)$

C $2(x-3)^2$ **D** $(2x+3)(x-6)$

8. Factor: $x^2 - 2x - 24$.

A $(x+6)(x-4)$ **B** $(x-6)(x+4)$

C $(x-8)(x+3)$ **D** $(x-12)(x+2)$

9. Factor: $3x^2 + 10x - 8$.

A $(3x+2)(x-4)$ **B** $(3x+4)(x-2)$

C $(3x-2)(x+4)$ **D** $(3x-4)(x+2)$

10. Solve $x^2 = 7x$.

A $x = 7$ only **B** $x = 0$ or $x = 7$

C $x = \pm\sqrt{7}$ **D** $x = 0$ or $x = -7$

11. How many real roots does $2x^2 - 5x + 4 = 0$ have?

A two different rational roots **B** two different irrational roots

C one repeated real root **D** no real roots

12. What is the vertex of $y = x^2 + 6x + 5$?

A $(-3, -4)$ **B** $(3, -4)$

C $(-3, 5)$ **D** $(3, 32)$

13. For $2x^2 - 8x + 3 = 0$, what are the sum and the product of the roots?

A sum -4, product $\frac{3}{2}$ **B** sum 4, product 3

C sum 4, product $\frac{3}{2}$ **D** sum 8, product 3

14. Which equation has roots 4 and -3?

A $x^2 + x - 12 = 0$ **B** $x^2 - x + 12 = 0$

C $x^2 + 7x - 12 = 0$ **D** $x^2 - x - 12 = 0$

15. A school chooses 20 students at random from each grade level. Which sampling method is this?

A stratified **B** simple random

C systematic **D** cluster

16. Test scores are normal with $\mu = 72$ and $\sigma = 6$. About what percent of scores are between 60 and 78?

A 68% **B** 81.5%

C 95% **D** 47.5%

17. The mean of a data set is 52 and the median is 40. Which statement is true?

A It is skewed left; report the mean.

B It is skewed right; report the median.

C It is symmetric; the mean and median are equally good.

D It is skewed right; report the mean.

18. A random sample of 1,600 people is polled. What is the margin of error?

A ±4%
B ±1.25%
C ±0.025%
D ±2.5%

Part B · Free Response (44 points)

19. Exponents and polynomials (8 points)

(a) Simplify $\dfrac{(2a^3b^{-2})^3}{4a^5b^{-4}}$. Write your answer with positive exponents only. **(2 points)**

Work:

(b) Solve $2^{x+3} = 8^{x-1}$. **(2 points)**

Work:

(c) Use the Factor Theorem to show that $(x-3)$ is a factor of $p(x) = x^3 - 2x^2 - 5x + 6$. Then factor $p(x)$ completely. **(4 points)**

Work:

20. Solving quadratic equations (10 points)

Use the method named in each part.

(a) By factoring: $x^2 + 2x - 35 = 0$ **(2 points)**

Work:

(b) By taking square roots: $(2x-1)^2 = 25$ **(2 points)**

Work:

(c) By completing the square: $x^2 - 8x + 3 = 0$. Give exact answers. **(3 points)**

Work:

(d) By the quadratic formula: $3x^2 + 4x - 2 = 0$. Give exact answers in simplest radical form, then round each to two decimal places. **(3 points)**

Work:

21. Graphing a parabola, and a word problem (10 points)

Use the parabola $y = -x^2 + 2x + 8$ for parts (a) to (e).

(a) Does the parabola open up or down? Does it have a maximum or a minimum? **(1 point)**

(b) Find the axis of symmetry and the vertex. **(2 points)**

Work:

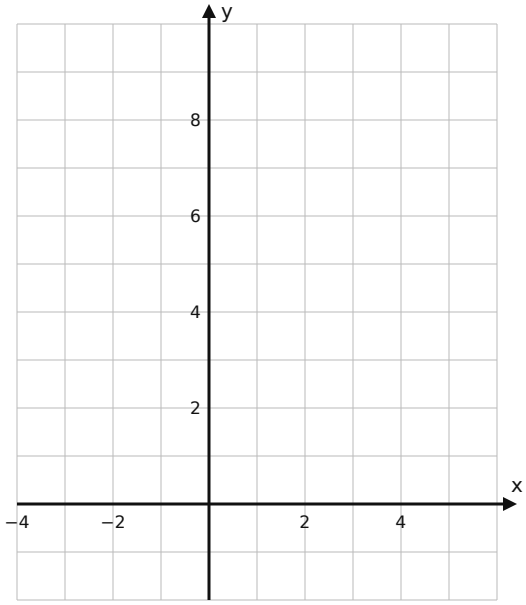
(c) Find the x-intercepts and the y-intercept. **(2 points)**

Work:

(d) Write the equation in vertex form and in factored form. **(1 point)**

Work:

(e) Sketch the parabola on the grid. Label the vertex and all three intercepts. **(1 point)**



(f) A ball is thrown straight up at 16 ft/s from a platform 32 ft high. Its height after t seconds is $h(t) = -16t^2 + 16t + 32$ feet. Find the maximum height of the ball, and the time when it hits the ground. **(3 points)**

Work:

22. Spread and the normal distribution (8 points)

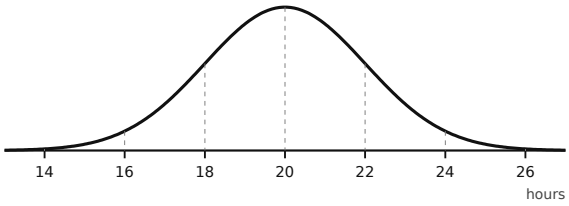
(a) Six students scored 4, 6, 7, 7, 8 and 10 on a quiz. Find the mean $\bar{x}$ and the standard deviation σ. Complete the table and round σ to two decimal places. **(3 points)**

x	$x - \bar{x}$	$(x - \bar{x})^2$
4		
6		
7		
7		
8		
10		
Sum		

Work:

The battery life of a phone model is normally distributed with mean $\mu = 20$ hours and standard deviation $\sigma = 2$ hours.

(b) Use the empirical rule to find the percent of phones whose battery lasts between 18 and 26 hours. Shade the region on the curve. **(2 points)**



Work:

(c) Find the z-score of 23.5 hours. Then use the table to find the percent of phones whose battery lasts longer than 23.5 hours. **(3 points)**

Work:

23. Collecting data and drawing conclusions (8 points)

(a) A survey of 250 of a school's 1,800 students finds that 34% walk to school. Identify the population and the sample. Is 34% a parameter or a statistic? Explain. **(2 points)**

(b) To find out how much students exercise, a researcher stands at the gym entrance after school and asks the first 50 students who walk in. Name the sampling method, explain why it is biased, and describe a better method. **(2 points)**

(c) In a random sample of 400 voters, 56% support building a new park. Find the margin of error and the interval that likely contains the true percent. Can the town conclude that a majority of voters support the park? Explain. **(2 points)**

Work:

(d) Marcus claims his coin is fair. He flips it 40 times and gets 27 heads. In 200 simulations of 40 flips of a fair coin, 27 or more heads came up 4 times. Using a 5% significance level, should he reject the claim that the coin is fair? Explain. **(2 points)**

Work: